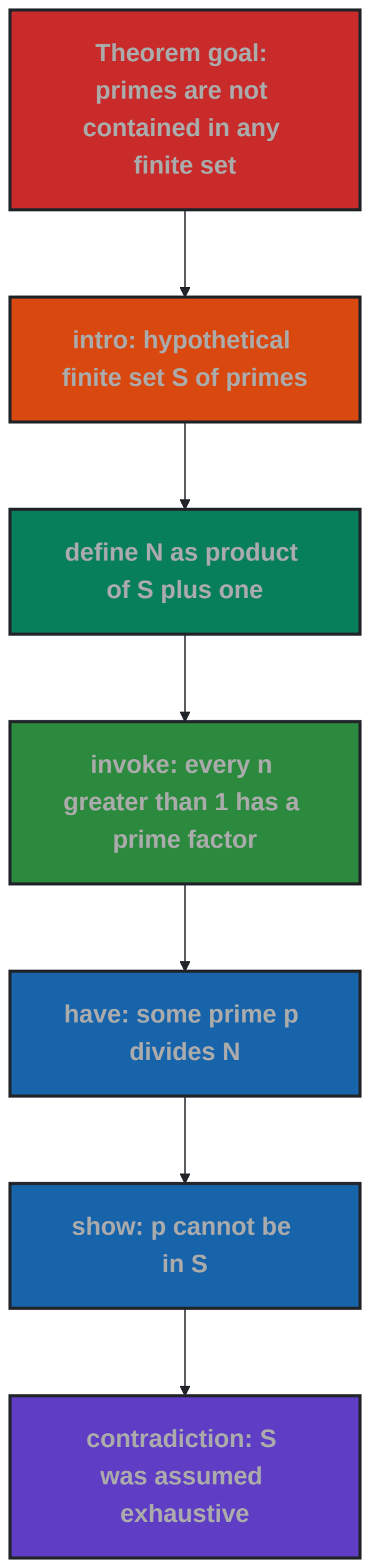


Theorem goal:  
primes are not  
contained in any  
finite set



```
graph TD; A["Theorem goal:  
primes are not  
contained in any  
finite set"] --> B["intro: hypothetical  
finite set S of primes"]; B --> C["define N as product  
of S plus one"]; C --> D["invoke: every n  
greater than 1 has a  
prime factor"]; D --> E["have: some prime p  
divides N"]; E --> F["show: p cannot be  
in S"]; F --> G["contradiction: S  
was assumed  
exhaustive"];
```

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finite set  $S$  of primes

define  $N$  as product  
of  $S$  plus one

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greater than 1 has a  
prime factor

have: some prime  $p$   
divides  $N$

show:  $p$  cannot be  
in  $S$

contradiction:  $S$   
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